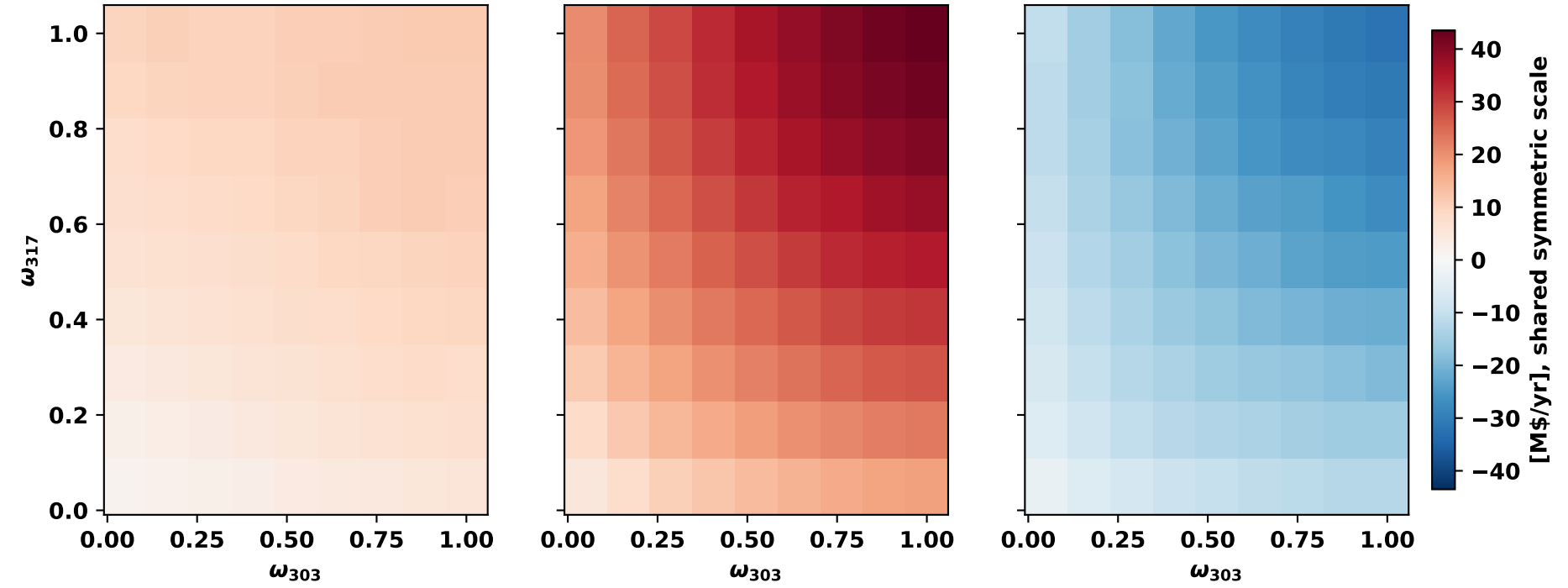


Cost decomposition, scenario A ($\rho_{H2} = 1.0$ \$/kg): the APPROX-column sign structure is raw minus synthetic bid cost
raw Δ cost = total - base **K_{syn} (synthetic bid cost)** **APPROX = raw - K_{syn}**



Identity check: max |raw - K_{syn} - less-synthetic| = 5.96e-08 USD over 81 runs (holds to < 1 USD). Raw cost is positive in every run; negative APPROX-column values arise from the principled synthetic-bid subtraction (19b SB1) — report them only as APPROX-column statements with this decomposition displayed.